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DAT540 - Introduction to Data science

Forecasting Daily Household Energy Consumption

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1 Introduction

Electrical power consumption plays a crucial role in optimizing power production, improving efficiency in modern households, and facilitating distribution. With the importance of green solutions, accurate short-term forecasting has become essential for consumers and producers to manage demand, enhance sustainability, and reduce costs.

1.1 Problem definition

In this project, we analyze a dataset of household consumption [10] that covers the period from December 2006 to November 2010. The dataset consists of 1-minute measurements of household electrical power consumption, containing nine columns: date, time, power consumption, voltage, current intensity, and sub-metered power. The idea is to use the data and predict the power consumption and visualize trend in the future using different models.

2 Methodology

Before using the dataset, it has to be cleaned and preprocessed, then divided into hourly and daily intervals for use in the different prediction models. The models are going to be plotted and the results compared. Different models prediction will be compared with the actual power consumption for a given time period to determine the accuracy of each model.

2.1 Key preprocessing

There were four key processing steps performed to prepare the dataset for time-series analysis. First, we handled missing values by converting '?' strings to NaN, dropped rows containing missing values, and forced all numeric columns to proper numeric types. Second, to enable accurate resampling, the date and time where merged into DateTime index. Third, the original minute based measurements were aggregated into daily averages, providing a stable time-series suitable for ARIMA modelling and moving-average. Finally, there where added a time-based feature for month and DayOfWeek to support trends and seasonal trends.

2.2 Models used

2.2.1 Moving Average model

The Moving Average (MA) model was chosen as a simple and transparent baseline for forecasting energy consumption. It predicts future values by averaging a fixed number of past observations, making it easy to interpret and useful for capturing short-term trends [4]. The moving average model also smooths the high-frequency noise in the time series, which is beneficial for the fluctuations in the household energy data file. In general moving average can be mathematically expressed as in equation 2.1

$$X_t = c + \epsilon_t + \theta_1 \cdot \epsilon_{t-1} + ... + \theta_n \cdot \epsilon_{n-1} \tag{2.1}$$

, where ϵ_n are the white noise terms associated with the time series and θ_n are the moving average constants. The interpretation of the MA model in practice uses the last parameter in ARIMA framework (see Chapter 2.2.3).
As a baseline model, MA provides a good reference for evaluating performance of more advanced models that will also be tested. It's main limitations is that it cannot model long-term trends or seasonal trends, making it most suitable for short-term forecasting tasks.

2.2.2 Exponential Smoothing

Exponential Smoothing is a forecasting method that generates predictions by giving more weight to recent observations and exponentially less weight to older ones. This allows the model to react smoothly to changing patterns without overreacting to noise [6]. The Holt-Winters method [11] consists of four main components:

- **Level (l)** – the baseline value at the current time
- **Trend (b)** – the direction and rate of change in the level
- **Seasonality (s)** – the repeating pattern at regular intervals
- **Smoothing parameters (α, β, γ)** – control how quickly the model reacts to new data

The fitting process estimates the smoothing parameters as well as the level, trend and seasonality parameters using numerical optimizers [7]. Holt–Winters captures level, trend, and seasonality with a simple, fast, and interpretable model that performs well for stable, regularly repeating patterns. But it struggles with sudden structural changes, irregular seasonality, and complex non-linear dynamics, and it requires consistent seasonal cycles to work effectively. It is well-suited for power consumption data because the series typically shows clear daily seasonal patterns and relatively smooth trends, which align exactly with the structures Holt–Winters is designed to model [8].

2.2.3 ARIMA/SARIMA

Autoregressive Integrated Moving Average combines Autoregressive Modelling (AR), differencing (I) and Moving Average (MV) components [9]. The AR part models how the current value depends on the previous value, mathematically in equation 2.2.

$$y_t = C + \phi_1 y_{t-1} + \phi_2 y_{t-2} + ... + \phi_n y_{t-n} \tag{2.2}$$

The I part removes the trend by differencing the series, mathematically in equation 2.3.

$$y'(t) = y_t - y_{t-1} \tag{2.3}$$

Moving average models the influence the past forecast errors (see Chapter 2.2.1 for further explanation). An ARIMA model uses the parameters ARIMA(p, d, q):

- p number of autoregressive lags
- d degree of the differencing required to make the series trend or stationary
- q number of moving-average terms on past forecast errors

SARIMA extends the ARIMA by adding seasonal components for better seasonal predictions [5] by introducing regular intervals like yearly, monthly and daily. SARIMA is represented as SARIMA$(p, d, q)(P, D, Q, s)$, where the first three parameters is the same as ARIMA, and the last four:

- P seasonal autoregressive order
- D seasonal differencing order
- Q Seasonal moving average order
- s length of the seasonal period

In practice, ARIMA captures overall trend, short-term dependences and is a strong baseline for compare against more complex models. While SARIMA captures in addition the repeating of seasonal cycles which is suitable for household energy consumption.

2.2.4 LSTM

Long Short-Term Memory (LSTM) can capture long-term dependencies in sequential data by using a memory cell holding information. LSTM selectively retain information that flows though the network and has a hidden state that acts like a short-term memory [3]. LSTM uses input, forget, and output gates to control how much information enters, stays in and leaves the memory cell. This allows the model to avoid the gradient problem and maintain long-term memory. The architecture of LSTM is a chain structure that contains neural networks and different memory blocks called cells. The upside of LSTM is that is uses non-linear relationships, long-term patterns and complex dynamics that less complex models like ARIMA/SARIMA and Exponential Smoothing cannot capture. However, LTSMs require more data, more computation and careful tuning of hyper parameters which make them more time consuming.

2.2.5 CNN–LSTM Model

The CNN–LSTM model combines convolutional layers with LSTM networks to capture both short-term and long-term patterns in time-series data [1]. The CNN extracts local features and fast fluctuations in the energy signal, while the LSTM learns how these features evolve over time and reflect recurring daily or weekly patterns. We used this model because hourly energy consumption contains both quick variations within each day and longer temporal dependencies. By combining CNN and LSTM, the model can represent these two types of patterns more effectively than either model alone, which often leads to improved forecasting accuracy in energy-related applications.

3 Results & Insights

This chapter presents the forecasting performance of all models applied to the household energy dataset. The models were trained on preprocessed historical data of 2 years and evaluated against unseen observations to assess their predictive accuracy. Performance is compared primarily using *root mean square error* (shown in table 1), allowing us to identify which models best capture the underlying consumption patterns and which struggle with variability and peak demand.

Models used	**RMSE (kW)**
Moving average	0.720
Exponential Smoothing	0.732
ARIMA/SARIMA	0.791
LSTM	0.808

Table 1: Root mean square error compresent for the different models used

The Time-of-Day **Moving Average** model produced an MAE of 0.525 kW and an RMSE of 0.72 kW on the test period. The model successfully captured the repeating daily pattern in the household's consumption.

Exponential Smoothing achieved an MAE of 0.5242 kW and an RMSE of 0.732 kW. Its performance aligns closely with the Moving Average model, although it remains less responsive to sudden consumption spikes.

The **ARIMA** model achieved an RMSE of 0.791 kW and an MAE of 0,55 kW, showing solid baseline performance. It captured overall trends well but was less effective during sharp peaks in consumption.

The **LSTM** model achieved an RMSE of 0.808 kW, showing that while it captured some general patterns in the household's energy consumption, it struggled with accuracy. It also struggled with the extreme peaks and learned instead the average behavior.

4 Conclusion

The moving average model provided a simple, but effective baseline for forecasting household energy consumption. By using a time of day approach, it successfully captured the daily consumption pattern and delivered stable predictions with reasonable accuracy. However, the model consistently smoothed out fluctuations and under predicted sudden peaks, limiting its usefulness for highly dynamic periods. Overall, it served as a reference point for the relic and demonstrated that incorporating daily seasonable can noticeably improve basic forecasting performance.

Exponential Smoothing is a valuable baseline approach for household power consumption forecasting. Holt-Winters with 24-hour seasonality achieved superior performance (MAE: 0.820 kW) by capturing critical daily patterns. Rolling window evaluation yielded MAE of 0.524 kW over 48-hour horizons with consistent performance. However, MAPE of $\sim$50% indicates magnitude accuracy limitations due to missing contextual features. The model requires around 5000 entries of historical data and 12 or 24 hour seasonal periods for optimal results. While excellent for lightweight operational forecasting, it should be combined with ensemble or deep learning methods for production-grade (10–15% MAPE) utility systems.

Using a 14-day rolling training window and generating 24-hour forecasts, the SARIMA model achieved RMSE: 0.7908 kW and MAE: 0.5501 kW. The model consistently captured daily cyclical behaviour and produced smooth forecasts with stable error levels across the three-month evaluation period. However, the gap between MAE and RMSE indicates that SARIMA struggled with sudden consumption spikes, leading to occasional large errors and wider confidence intervals during high-variability periods. Although effective for short-term operational forecasting and well suited for data with strong regular seasonality, the model remains limited by its linear structure and absence of contextual features. For production-grade forecasting systems, SARIMA should therefore be combined with exegenous variables or integrated into ML-based pipelines to better capture nonlinear household behaviour.

Studies have shown that LSTM models perform better when the time-series data is over multiple years. For example, Yu et al. (2021)[12] used five years of weather-station data and found significant accuracy improvement when the neural network was trained on the extended history. Similarly, Bulut (2021) [2] used 10-12 years of hydro-generation data to predict future production and achieved favourable results. In contrast, our model was trained on 2 years of data, which is far fewer seasonal cycles and might explain the relatively poor performance.

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